

TELCO OFFER, LOAD & TRANSACTION DATA ANALYSIS WITH SQL

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Which Product is most ordered per bundle type?

```
WITH BOX AS
(SELECT
BUNDLE_TYPE,
CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
COUNT(ORDER_ID) AS TOTAL_ORDER,
ROW_NUMBER() OVER (PARTITION BY BUNDLE_TYPE ORDER BY COUNT(ORDER_ID) DESC) AS
RANKS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY 1,2)
SELECT * EXCEPT(RANKS)
FROM BOX WHERE RANKS=1;
```

Which Bundle Type is least ordered in each Division?

```
WITH Y AS
(SELECT
BUNDLE_TYPE,
DIVISION,
CONCAT(UPPER(OPERATOR), "- ", PRODUCT_NAME) AS PRODUCT_NAME,
COUNT(ORDER_ID) AS TOTAL_ORDER,
ROW_NUMBER() OVER (PARTITION BY DIVISION ORDER BY COUNT(ORDER_ID) DESC) AS
RANKS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1,2,3)
SELECT * EXCEPT(RANKS)
FROM Y
WHERE RANKS = 1;
```

How Many Products are there per Bundle Type?

```
SELECT
BUNDLE_TYPE,
COUNT(DISTINCT PRODUCT_ID) AS TOTAL_PRODUCTS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1;
```

What is the total revenue and average selling price per product, bundle type, operator and division?

```
SELECT
    DIVISION,
    OPERATOR,
    SUM(CASE WHEN BUNDLE_TYPE = 'Internet' THEN SELLING_PRICE ELSE 0 END) AS
TOTAL_REVENUE_INTERNET,
    AVG(CASE WHEN BUNDLE_TYPE = 'Internet' THEN SELLING_PRICE ELSE 0 END) AS
AVG_SELLING_PRICE_INTERNET,
    SUM(CASE WHEN BUNDLE_TYPE = 'Intenet & Minute' THEN SELLING_PRICE ELSE 0
END) AS TOTAL_REVENUE_INTERNET_AND_MINUTE,
    AVG(CASE WHEN BUNDLE_TYPE = 'Intenet & Minute' THEN SELLING_PRICE ELSE 0
END) AS AVG_SELLING_PRICE_INTERNET_AND_MINUTE,
    SUM(CASE WHEN BUNDLE_TYPE = 'Minute' THEN SELLING_PRICE ELSE 0 END) AS
TOTAL_REVENUE_MINUTE,
    AVG(CASE WHEN BUNDLE_TYPE = 'Minute' THEN SELLING_PRICE ELSE 0 END) AS
AVG_SELLING_PRICE_MINUTE
FROM
    `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
WHERE
    DIVISION IS NOT NULL
GROUP BY
    1, 2;
```

Which product had the highest number of sales and what was the total revenue for that product?

```
SELECT
    CONCAT (UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
    COUNT(ORDER_ID) AS COUNTS,
    SUM(SELLING_PRICE) AS TOTAL_REVENUE
FROM
    `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY 1
ORDER BY COUNTS DESC
LIMIT 1;
```

Find the Product and Bundle Type Contributing the Highest Percentage of total Revenue?

```
WITH X AS (
    SELECT
        BUNDLE_TYPE,
        CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
        SUM(SELLING_PRICE) AS TOTAL_REVENUE,
        ROUND(SUM(SELLING_PRICE) / (SELECT SUM(SELLING_PRICE) FROM `organic-reef-427307-i3.offer_load.Offer_load_Transaction`) * 100, 2) AS PERCENTAGE,
        ROW_NUMBER() OVER (ORDER BY SUM(SELLING_PRICE) DESC) AS RANKS
    FROM
        `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
    GROUP BY
        BUNDLE_TYPE, OPERATOR, PRODUCT_NAME
)
```

```
SELECT *
FROM X
WHERE RANKS = 1;
```

What is the Maximum and Minimum amount of Discount and commission per product?

```
SELECT
CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
MAX(commission) AS MAX_COMMISSION,
MIN(commission) AS MIN_COMMISSION,
MAX(discount) AS MAX_DISCOUNT,
MIN(discount) AS MIN_DISCOUNT
FROM
`organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY 1;
```

Which Operator, Division, offers the Highest avg discount across Products?

```
SELECT
UPPER(OPERATOR) AS OPERATOR,
UPPER(DIVISION) AS DIVISION,
ROUND(AVG(DISCOUNT), 2) AS AVG_DISCOUNT
FROM
`organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY
1, 2
ORDER BY AVG_DISCOUNT DESC
LIMIT 1;
```

How Many Unique Customers have placed orders and how many are repeat buyers?

```
SELECT
CUSTOMER_ID,
COUNT(DISTINCT DATE(ORDER_TIME)) AS ORDER_COUNT
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY 1
HAVING ORDER_COUNT > 1;
```

Which customer has the highest number of orders or total spending?

```
SELECT
CUSTOMER_ID,
COUNT(ORDER_ID) AS TOTAL_ORDERS,
SUM(SELLING_PRICE) AS TOTAL_SPENDING
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1
order by
TOTAL_ORDERS DESC
LIMIT 1;
```

What is the avg purchase frequency of customers and how does it vary by bundle type.

```
WITH CTE AS
(SELECT
BUNDLE_TYPE,
CUSTOMER_ID,
COUNT(ORDER_ID) PURCHASE_FREQUENCY
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY 1,2)
SELECT
BUNDLE_TYPE,
CASE WHEN CTE.PURCHASE_FREQUENCY BETWEEN 1 AND 5 THEN '1-5'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 6 AND 10 THEN '6-10'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 11 AND 15 THEN '11-15'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 16 AND 20 THEN '16-20'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 21 AND 25 THEN '21-25'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 26 AND 30 THEN '26-30'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 31 AND 35 THEN '31-35'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 36 AND 40 THEN '36-40'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 41 AND 45 THEN '41-45'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 46 AND 50 THEN '46-50'
ELSE 'ABOVE 50' END AS FREQUENCY_BIN,
COUNT(DISTINCT CUSTOMER_ID) AS COUNT,
ROUND(AVG(PURCHASE_FREQUENCY),2) AS AVG_PURCHASE
FROM
CTE
GROUP BY 1,2;
```

During which time of the day, most orders are placed?

```
SELECT
EXTRACT(HOUR FROM ORDER_TIME)AS ORDER_HOUR,
COUNT(ORDER_ID) AS TOTAL_ORDERS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1
ORDER BY
TOTAL_ORDERS DESC
LIMIT 1;
```

What is the average fulfillment or delivery time between order time and delivery for each operator and division?

```
SELECT
UPPER(OPERATOR) AS OPERATOR,
ROUND(AVG(DATE_DIFF(DELIVERY_TIME, ORDER_TIME, MINUTE)),2)AS DELIVERY_TIME
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
ORDER_STATUS = "Confirm"
GROUP BY 1;
```

Calculate the average delivery time for orders in each division.

```
SELECT
UPPER(DIVISION),
ROUND(AVG(DATE_DIFF(DELIVERY_TIME, ORDER_TIME, MINUTE)),2)AS DELIVERY_TIME
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
ORDER_STATUS = 'Confirm'
GROUP BY
1;
```

What are the cancellation rate and complaint rate per operator, and division?

```
WITH X AS
(SELECT
DIVISION,
OPERATOR,
COUNT(ORDER_ID) AS TOTAL_ORDERS,
COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Cancel" THEN ORDER_ID ELSE NULL
END) AS CANCELLED_ORDERS,
COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Complain" THEN ORDER_ID ELSE NULL
END) AS COMPLAINT_ORDERS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE DIVISION IS NOT NULL
GROUP BY
1,2)
SELECT
DIVISION,
OPERATOR,
ROUND(CANCELLED_ORDERS/ X.TOTAL_ORDERS* 100,2) AS CANCEL_PERCENTAGE,
ROUND(COMPLAINT_ORDERS/ X.TOTAL_ORDERS* 100,2) AS COMPLAINT_PERCENTAGE
FROM
X;
```

What are the most common reasons for complaints?

```
SELECT
REASON,
COUNT(REASON) AS MOST_COMMON_REASON
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1
ORDER BY MOST_COMMON_REASON DESC
LIMIT
1;
```

How many orders were placed in each division?

```
SELECT
  DIVISION,
  COUNT(ORDER_ID) AS TOTAL_ORDERS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE DIVISION IS NOT NULL
GROUP BY
  1;
```

Which division has the highest number of total sales and transactions?

```
SELECT
  DIVISION,
  SUM(SELLING_PRICE) AS TOTAL_SALES_VALUE,
  COUNT(ORDER_ID) AS TOTAL_TRANSACTIONS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
  DIVISION
ORDER BY
  TOTAL_SALES_VALUE DESC
LIMIT 1;
```

What is the percentage of order status (e.g., completed, canceled) per operator and division?

```
WITH Y AS
(SELECT
  DIVISION,
  OPERATOR,
  COUNT(ORDER_ID) AS TOTAL_ORDERS,
  COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Confirm" THEN ORDER_ID ELSE NULL
END ) AS TOTAL_CONFIRMED,
  COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Cancel" THEN ORDER_ID ELSE NULL
END) AS TOTAL_CANCELED
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
  DIVISION IS NOT NULL
GROUP BY
  1,2)
SELECT
  UPPER(DIVISION) AS DIVISION,
  UPPER(OPERATOR) AS OPERATOR,
  ROUND( TOTAL_CONFIRMED/Y.TOTAL_ORDERS * 100,2) AS COMPLETION_PERCENTAGE,
  ROUND(TOTAL_CANCELED/ Y.TOTAL_ORDERS * 100,2) AS CANCELETION_PERCENTAGE
FROM
  Y;
```

How many products are offered by each operator?

```
SELECT
UPPER(OPERATOR) AS OPERATOR,
COUNT(DISTINCT PRODUCT_ID)AS TOTAL_PRODUCTS
FROM
    organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
    1;
```

Calculate product popularity by validity period.

```
SELECT
    VALIDITY,
    COUNT(ORDER_ID) AS POPULARITY
FROM
    organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
    1
ORDER BY
    POPULARITY DESC;
```

-- ADVANCED CODE USING CTE AND WINDOWS FUNCTION

```
WITH M AS (
    SELECT
        VALIDITY,
        CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
        COUNT(ORDER_ID) AS POPULARITY,
        ROW_NUMBER() OVER (PARTITION BY VALIDITY ORDER BY COUNT(ORDER_ID) DESC)
    AS POPULARITY_BY_VALIDITY
    FROM
        `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
    GROUP BY
        VALIDITY, OPERATOR, PRODUCT_NAME
)
SELECT
    VALIDITY,
    PRODUCT_NAME,
    POPULARITY,
    POPULARITY_BY_VALIDITY
FROM
    M
ORDER BY
    POPULARITY DESC;
```

Identify products with the most complaints and cancellations.

```
SELECT
    CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
    COUNT(COMPLAIN_STATUS) AS TOTAL_COMPLAINTS,
    COUNT(CASE WHEN ORDER_STATUS = "Cancel" THEN ORDER_ID ELSE NULL END) AS
    TOTAL_CANCELLATIONS
FROM
```

```

    organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
    1
ORDER BY
    TOTAL_COMPLAINTS DESC,
    TOTAL_CANCELLATIONS DESC
LIMIT
    1;

```

Calculate Potential revenue loss due to cancellation

```

SELECT
    SUM(SELLING_PRICE) AS TOTAL_REVENUE_LOSS,
FROM
    organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
    ORDER_STATUS = "Cancel";

```

What are the top 3 or 5 most-selling products or bundles in key divisions like Dhaka and Rangpur?

```

WITH Z AS
(
    SELECT
        DIVISION,
        PRODUCT_NAME,
        BUNDLE_TYPE,
        COUNT(ORDER_ID) AS TOTAL_SELLING,
        ROW_NUMBER() OVER (PARTITION BY DIVISION ORDER BY COUNT(ORDER_ID) DESC) AS
RANKS
    FROM
        organic-reef-427307-i3.offer_load.Offer_load_Transaction
    WHERE
        DIVISION IN ('Dhaka', 'Rangpur')
    GROUP BY
        1, 2, 3
    ORDER BY
        DIVISION, TOTAL_SELLING DESC
)

SELECT
    DIVISION,
    PRODUCT_NAME,
    BUNDLE_TYPE,
    Z.TOTAL_SELLING
FROM
    Z
WHERE
    RANKS <= 5;

```